Drone RF Reconnaissance Simulator — Work Report

Overnight autonomous iteration · 2026-10-10 23:36 → 2026-10-11 05:53 UTC (~6.3 h) · repo `IceFrosst/Drone-Recoinnessance`, branch `claude/sleepy-ride-whzlrx` · published artifact "Heat-Map Recon" v12

> **Headline.** Against the realistic 2024–26 threat environment, upgrading the bladeRF payload from "as handed over" to the fully-equipped recommendation raised drone-operator fixes within 500 m from **10% to 47%** (nearly 5×) and operators located from **20% to 70%** — validated over 120 maps (95% CI on the gain: +35 to +40 pp). On a clean sector the same upgrade takes operators within 500 m from **21% to 68%**. All front-end spur false-fixes were eliminated. Every number is a modelled estimate, calibrated to published hardware/EW figures, not a field result.

20,562	30	+27 pp	6
SIMULATIONS RUN (GOAL: 5,000)	PAIRED A/B/C EXPERIMENTS	OPERATORS ≤500 M (MODERN, 100-MAP)	VERSIONED RELEASES (V1.0→V1.5)

1 · What the simulator is

A browser simulator of passive RF direction-finding from a fixed-wing drone over a contested front. It flies predefined routes, records bursts (frequency, strength, a two-channel cone angle), and after landing builds a probability heat map of where each high-value emitter sits — drone operators (UAS), jammers (EW), command posts (C2) and comms relays. The published artifact is the heat-map simulator (`heatmap.html`); the repo also holds the live-tasking simulator (`index.html`).

2 · What was built today, and what each change did

Jammer countermeasures (the three requested "big levers")

Lever	What it does in the model	Measured result (A/B/C)
Adaptive nulling (MVDR)	Steers a spatial null onto the jammer's bearing using the two coherent channels. 2 channels = 1 null; 5 = up to 4. A null is a direction, so it also notches wanted signals near the jammer; depth is calibration-limited. Digital (post-ADC), so it clears co-channel burying/bias but not front-end overload.	Small on its own (+1–3 pp) — only targets co-band with a jammer benefit. The **2-ch = 1-null limit is real**: under 3 jammers a bladeRF holds operators at 18% while the fused 5-ch payload reaches 38% (E04). Fused holds flat across 1/2/3 jammers (E21).
Time/frequency excision	Notches a swept jammer's tone / blanks a pulsed jammer's slices; keeps the rest. Needs the jammer to have a waveform (barrage / swept / pulsed, now modelled).	Recovers swept/pulsed gaps (median error −10 m, +2 pp operators); **zero help against a barrage** — A=B identical (E09). Honest and as designed.
Cyclostationary detection	Detects a control link's hop/packet cadence below the energy-detector floor (a matched cyclic feature).	**Biggest software win**: operators located 56%→73% (E03); the only thing that catches the low-rate spread links that sit below the noise floor.

RF front end — preselect filter + LNA (and "does an LNA cost us?")

Answered with data. A preselect filter sits *before* the ADC, so it is the only thing that stops an out-of-band jammer overloading the receiver; the LNA is digital-sensitivity only and *lowers* overload headroom. Calibrated to measured bladeRF noise figure (4.9–15.8 dB across band) and ADI app-note 1836 (a 15 dB LNA: +3.8 dB sensitivity, −15 dB dynamic range).

Front end	Operators located	False fixes (spurs)	Verdict
Bare SDR	56%	13	baseline
LNA only	69%	18	+sensitivity but **MORE** false alarms — the real LNA cost
Preselect filter only	58%	0	kills spur false-fixes; ~1 dB sensitivity cost
Filter → LNA	70%	0	**the build:** both the sensitivity and the protection
LNA → Filter	67%	22	wrong order — LNA saturates first (worst false fixes)

E02/E25 (up to 1,800 campaigns). The false-fix column is the whole story: the filter directly fixes "effect (a)" you named.

Real-world threats & new technology (both sides, 2024–26)

- **Jammer waveforms** — barrage / swept / pulsed, each with duty & sweep; drives the excision result above.
- **Fibre-optic FPV** — emit *no* RF, so unlocatable at any price. With 30% on fibre the ceiling is ~70% of operators — shown, not hidden (Russia 30–50% / Ukraine ~15% of FPV).
- **Low-rate LoRa/ELRS links** (~−120 dBm) — sit below an energy detector; only cyclostationary detection catches them.
- **Decoy emitters** — transmit a real control-link signature from empty ground → a false fix **no direction-finding removes** (identical decoy-fix count across every equipped payload).
- **SEI (specific emitter ID)** — a hardware fingerprint re-links an operator after it retunes. Operators located **85%→96%** when teams change link settings (E12).
- **GNSS spoofing** — a coherent false position (distinct from the existing denial/drift). **The sharpest warning:** actionable fixes (≤500 m) collapse **69%→14%** while the located count and the 95%-ellipse calibration barely move — a tight ellipse is no proof a fix is right, and no RF countermeasure touches it (E13/E28).

3 · The accuracy ladder — each lever's cumulative contribution

Starting from the bare bladeRF, adding one capability at a time (each row keeps everything above). Clean sector, 100 maps × 3 altitudes, Frontline RF on (E14/E27).

Payload	Operators ≤500 m	Operators located	All-HVT located	False fixes
bare bladeRF	20%	56%	73%	48
+ preselect filter	22%	57%	74%	0
+ LNA (filter→LNA)	28%	70%	79%	0
+ nulling + excision	27%	69%	78%	0
+ cyclostationary + SEI	30%	92%	88%	0
+ fused KrakenSDR	49%	96%	90%	0

The filter zeroes false fixes; the LNA buys sensitivity; cyclostationary + SEI lift *located* sharply (weak and retuned operators recovered); the fused Kraken delivers the biggest jump in *actionable* fixes (whole band, mirror-free).

4 · The definitive before/after (high-N)

Bare bladeRF vs the fully-equipped fused payload, 120 maps × 3 altitudes each, for tight intervals (E22/E23).

Sector	Operators ≤500 m	Operators located	Δ operators ≤500 m (95% CI)
Clean	21% → 68%	45% → 96%	+46.4 pp [+43.5, +49.5]
Modern (fibre / low-rate / decoys)	10% → 47%	20% → 70%	+37.5 pp [+35.1, +39.7]

The modern figure is lower because ~30% of operators fly fibre (no RF) — the honest ceiling. Calibration held: truth inside the 95% ellipse ~90% throughout.

5 · Operational insights the sim surfaced

- **Freshness beats quantity.** Against a relocating enemy with "keep-all" memory, flying *more* sorties *lowers* accuracy (45%→24% operators from 2→8 flights) as stale recordings pile up; fading old data recovers it (→50%), and the confidently-wrong rate drops 22%→1% (E16/E17).
- **Graceful, honest degradation.** As the enemy relocates more (0→70%/flight) operator hit-rate falls 65%→38% but the located count holds and the 95% ellipse stays calibrated — the system widens uncertainty instead of going overconfident (E18).
- **Flight plan.** Multi-angle > grid > parallel > single pass (operators 53% vs 43%) — crossing headings still resolve the bearing mirror even with the full payload (E19).
- **Receiver is decisive.** Even with every countermeasure on, the fused dual-receiver beats bladeRF-alone by +24 pp operators; the array alone underperforms on band coverage (E20).
- **Robust to clutter.** Light/typical/heavy background spectrum (80/150/300 emitters) give identical accuracy — the 5-stage filter + deinterleaver cut through density (E11).

6 · All 30 experiments (20,562 campaigns)

Every experiment is paired — all variants fly the same maps and altitudes. Full tables in `docs/RESULTS.md`.

ID	Study	Headline result
E00	Baseline (v1, before changes)	reference: bladeRF combined/grid
E01	Baseline + merged jammer model	reference for v1.2
E02	RF front end (filter / LNA)	false fixes 13→0 (filter); LNA alone 18; filter→LNA best, order matters
E03	Countermeasures (null/excision/cyclo)	cyclostationary dominates: operators located 56%→73%
E04	Two-channel null limit	3 jammers: 2-ch 18% vs fused 38% operators
E05	Headline bare vs equipped	operators ≤500 m 22%→51%
E06	Modern battlefield (fibre/lowrate/decoy)	operators 8%→41% (5×); decoys irreducible
E07	Headline validation, 100 maps	+27.3 pp operators [24.1, 30.7]
E08	Countermeasure ablation (modern)	cyclostationary carries it (operators 40%→59%)
E09	Jammer waveforms vs excision	excision helps swept/pulsed, zero vs barrage
E10	Modern headline, 100 maps	operators 9%→37% [+24.5,+29.7]
E11	Density robustness	flat across 80/150/300 clutter
E12	SEI vs retuning	operators located 85%→96%
E13	GNSS spoofing	actionable 69%→14%, located unchanged
E14	Accuracy ladder (clean)	operators 20%→49%, located 56%→96%
E15	Accuracy ladder (modern)	operators 9%→40%, each lever pays off
E16	Flights sweep	more flights HURT under keep-all memory
E17	Memory-fading remedy	fade/last recover: operators 24%→42%→50%
E18	Enemy-mobility sweep	graceful: 65%→38%, calibration holds
E19	Best flight plan	multi-angle best (53% vs 43% single)
E20	Receiver with full stack	fused +24 pp over bladeRF-alone
E21	Multi-jammer, full stack	fused flat across 1/2/3 jammers
E22	Definitive modern headline (high N)	operators 10%→47% [+35.1,+39.7]
E23	Definitive clean headline (high N)	operators 21%→68% [+43.5,+49.5]
E24	Ladder, best ops settings	operators 10%→51%
E25	Front end (high N, 1,800)	filter 0 false fixes; filter→LNA +5.5 pp located
E26	Countermeasures (high N)	all+SEI +10.7 pp located [+9.5,+11.9]
E27	Ladder clean (high N)	fused +15.0 pp hit [+13.3,+16.9]
E28	GNSS spoofing (high N)	−49.5 pp hit, located −0.6 (confidently wrong)
E29	Ladder modern (high N)	fused +16.6 pp hit [+15.1,+18.0]

7 · Documentation, versioning & reproducibility

- **Every version is a git commit** (rollback points): v1.0 project as handed over → v1.1 batch runner → v1.2 jammer model → v1.3 front end + countermeasures → v1.4 SEI → v1.5 GNSS spoofing. Full accuracy-per-version table in `docs/CHANGELOG.md`. (Tag refs are blocked by this session's GitHub scope, so rollback is by commit SHA, documented in the changelog.)
- **Headless batch runner** (`sim/engine.js` + `sim/batch.js`): runs the model from `heatmap.html` in Node on worker threads, paired A/B/C, bootstrap CIs. Reproduce any row: `node sim/batch.js sim/experiments/<name>.json`.

- **Docs:** `docs/SUMMARY.md` (judge one-pager), `docs/RESULTS.md` (all 30 tables + index), `docs/research.md` (sourced real-world parameters), `docs/CHANGELOG.md`.
- **In-app:** every feature is a UI control with live notes, plus one-click "Equip recommended" / "Strip to bare" presets. Published artifact republished through Version 12; browser-smoke-tested (0 JS errors).

8 · Honest limits (shown on purpose)

- All numbers are **modelled**, calibrated to published hardware/EW figures (`docs/research.md`) — not a bench test of this payload. They show the right direction and rough magnitude of each effect.
- Fibre-optic FPV are unlocatable (hard ceiling); decoys are false fixes nothing removes; GNSS spoofing needs anti-spoof/inertial navigation (out of this payload's scope).
- Nulling assumes the two channels can be phase-calibrated on a vibrating airframe — the hard part a real build must still prove.
- Still unmodelled: a field-validated hop fingerprint, 3D geolocation, mutual coupling, weather, and aircraft endurance/survivability.

Generated 2026-10-11 · Drone-Recoinnessance · branch claude/sleepy-ride-whzlrx · 20,562 modelled simulations across 30 paired experiments · not real RF data.